MARGDARSHAK GWALIOR

BEYOND THE FORT | SOLUTION OVERVIEW & BASIC SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

1. Solution Overview

MargDarshak Gwalior ("Beyond the Fort") is a hybrid destination-intelligence engine designed to bridge the temporal and spatial coordination gap faced by short-stay and cultural visitors. Its sequential pipeline combines a deterministic constraint-and-routing engine with a grounded knowledge-retrieval layer. This ensures that time-constrained travellers receive mathematically feasible, culturally respectful, and economically inclusive itineraries.

2. System Actors (Users)

ACTOR	ROLE / NEED
Primary User — Transit Explorer	Domestic rail or highway travellers with a strict 4–8 hour window who need an optimized and feasible sightseeing route.
Secondary User — Cultural Heritage Seeker	Visitors interested in living music history, traditional stone crafts, and authentic local culinary heritage.
System Actor — Data Administrator	Heritage managers and local curators who verify and update opening hours, tariffs, and site rules.

3. Inputs and Outputs

SYSTEM INPUTS	SYSTEM OUTPUTS
Time budget: slider from 2.0 to 10.0 hours. **Origin:** Gwalior Junction Railway Station by default, with other transit nodes available. **Temporal context:** day of week and exact arrival time (e.g., Monday, 10:30 AM). **Interests:** Architecture, Hindustani Music, ODOP Crafts, and Street Food. **Mobility:** "Avoid Steep Stairs" toggle for elderly or mobility-impaired visitors.	**Sequenced itinerary:** arrival, visit, and departure times for each stop. **Route map:** turn-by-turn navigation and mode recommendations (auto-rickshaw, e-rickshaw, walking). **Cost breakdown:** itemized admission charges across agencies and estimated transit fares. **Micro-merchant inclusions:** verified craft workshops or traditional food stops within 800 m of heritage sites. **Contextual notes:** source-attributed historical summaries and site-specific etiquette guidance.

4. System Workflow

STEP	PROCESS	DESCRIPTION
01	**Intent Parsing**	The UI captures constraints; AI/NLP converts natural-language or Hinglish requests into structured parameters.
02	**User Confirmation**	The user reviews and confirms the parsed parameters before route generation.
03	**Deterministic Filtering**	Boolean filters against the verified operational database eliminate attractions closed on the selected day or time.
04	**Topological Routing**	An elevation-aware distance matrix produces a feasible, low-backtracking route within the time budget.
05	**Contextual Enrichment**	Verified cultural notes, tariff details, and nearby micro-merchants are attached to the finalized route.

FUNCTIONAL & NON-FUNCTIONAL REQUIREMENTS

5. Functional Requirements (FR)

ID	REQUIREMENT	SPECIFICATION
FR-1	**Temporal Feasibility Filter**	Exclude attractions whose verified operating hours do not overlap with the scheduled visit window, including day-of-week closures.
FR-2	**Topological Route Optimization**	Minimize backtracking and distinguish vehicular-accessible points (e.g., Urvai Gate) from steep pedestrian paths (e.g., Hathi Pol).
FR-3	**Time Budget Guarantee**	Ensure the combined site-visit durations and transit times never exceed the user's stated time budget.
FR-4	**Transparent Tariff Breakdown**	Display verified admission prices and itemize fees by managing authority (ASI, State, or Private Trust).
FR-5	**Micro-Merchant Inclusion**	For each primary monument visited, highlight at least one verified local craft workshop or traditional food stop within 800 metres.
FR-6	**Verified Knowledge Retrieval**	Ground historical information in a verified document store of academic and official texts, with source attribution for factual claims.
FR-7	**Mobility Constraint Adherence**	When "Avoid Steep Stairs" is enabled, exclude pedestrian climbs and recommend vehicular drop-offs only where operationally feasible.
FR-8	**Source Citations**	Display the managing authority and the exact last-verification date for each attraction's operational details.

6. Non-Functional Requirements (NFR)

ID	REQUIREMENT	SPECIFICATION
NFR-1	**Performance**	The deterministic routing engine shall generate valid itineraries in under 1.5 seconds.
NFR-2	**Reliability**	The application shall operate using local or cached datasets when external connectivity is unavailable.
NFR-3	**Usability**	The web interface shall remain responsive on mobile browsers down to a 360 px viewport width.
NFR-4	**Transparency**	Information shall be labelled by source type: Official Fact, Operator Listing, or Calculated Estimate.
NFR-5	**Privacy**	No mandatory personal registration, background location tracking, or storage of user identity data.
NFR-6	**Graceful Degradation**	If the generative language model is unavailable, show pre-written, verified historical summary cards without interrupting core itinerary generation.

AI GOVERNANCE, CONSTRAINTS & ERROR HANDLING

7. AI vs. Human Responsibilities

RESPONSIBILITY LAYER	SCOPE	CONTROL / ACCOUNTABILITY
Deterministic Operations (No AI permitted)	Schedule validation; topological routing; tariff and cost arithmetic; time-budget validation; data-freshness checks.	Rule-based calculations and explicit constraints determine feasibility. Generative output cannot override these results.
AI and NLP Operations (Strictly constrained)	Parsing natural-language and Hinglish intent; summarizing verified historical texts; formatting the itinerary narrative.	AI may use only the local verified document store for historical summaries and must preserve source attribution.
Human-in-the-Loop	Administrators verify hours, tariffs, and merchant status; users confirm parsed inputs; specialists vet source texts; business owners confirm visitor acceptance.	Humans maintain the trusted data layer and confirm operational details that require local or domain knowledge.

8. Constraints and Error-Handling Protocols

SCENARIO	SYSTEM RESPONSE
Unreliable or stale data	If an operational record has not been updated within 90 days, display a “Requires Local Confirmation” badge alongside the ticket-window phone number.
Conflicting data / schedule clash	If a requested site is closed on the selected day (e.g., Gujari Mahal on Monday), block it from the route, cite the official schedule, and suggest an open alternative (e.g., Man Mandir Palace).
Missing data / infeasible route	If all requested sites are closed or exceed the time budget, return “No feasible stops found” and offer a simplified single-site itinerary instead of inventing a route.
AI hallucination guardrail	AI must not generate historical facts, opening hours, fares, or accessibility claims. It may retrieve and summarize verified documents only, with source attribution.

DESIGN PRINCIPLE
Feasibility is decided by deterministic rules; AI is limited to intent interpretation and grounded explanation; humans remain accountable for the accuracy and freshness of operational data.